

Invalid Is Not Failure: Infrastructure Censoring Bias in Long-Horizon Agent Evaluation

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Abstract

Long-running agents sometimes stop without a valid task outcome because the provider, runtime, or harness failed rather than the agent. Reported reliability then depends on a reporting choice: score such runs as failures, or drop them. We show these choices err in opposite directions, and that the error persists even when infrastructure failures are independent of agent behaviour: a per-step hazard compounds to an observation probability s^H in the horizon H , so censoring becomes informative whenever H is associated with success. We give assumption-free identification bounds, then show the complete-case estimand satisfies $p_{cc} - p = \text{Cov}(Y, s^H)/\mathbb{E}[s^H]$, while scoring invalid runs as failures can never overstate p . Two public corpora — 80,036 software-engineering and 4,950 tool-agent trajectories — behave exactly as the theory predicts: where successful runs are much shorter, dropping invalid runs inflates reliability by 2.7 points at a 2% hazard; where horizons are nearly outcome-independent, that bias almost vanishes while invalid-as-failure still costs 12.6 points. Censoring-aware weighting recovers the estimand when the mechanism is known. We recommend benchmarks report validity counts and censoring sensitivity rather than silently choosing a policy.

1 Introduction

A long-horizon agent run can end for two very different reasons. The agent can fail the task, or the measurement apparatus can fail the agent: a provider quota is exhausted, an API returns 503, a container dies, a harness times out. Only the first is evidence about the agent. Yet benchmark reports usually collapse both into one success rate through an unstated policy — either invalid runs are scored as failures, or they are deleted.

We are not the first to find validity problems in agent benchmarks (Kapoor et al. 2024; Weidinger et al. 2025; Zhu and Chang 2026), and a recent audit finds that agent benchmark papers disclose little about their harness, inference, and failure handling (Naser Moghadasi and Ghaderi 2026). Our claim is narrower: the two obvious ways of handling invalid runs are biased in opposite directions, and neither is conservative. The reason is structural. Suppose infrastructure failures are entirely independent of what the agent is doing, arriving with

a small constant hazard q per agent decision step. A run of horizon H then survives with probability s^H , $s = 1 - q$. Long runs are exposed more often, so an outcome-independent per-step failure becomes informative at the trajectory level as soon as horizon is associated with success. On our primary corpus resolved runs average 15.2 agent steps and unresolved runs 28.7.

Terminology. We use censoring operationally, to denote premature termination before a semantic endpoint is observed. Statistically this induces missing binary outcomes through a sequential observation process; we do not claim the task outcome is a conventional time-to-event variable, and H is a trajectory horizon rather than a clinical event time. We write π_i for an observation probability. Throughout we keep three run states distinct: a valid success, a valid task failure, and an infrastructure-invalid (censored) run whose semantic outcome was never observed. A provider or runtime failure is an instance of the third, never of the second.

Contributions.

- **Formal.** Assumption-free identification bounds for reliability under censoring, and a bias identity showing horizon converts outcome-independent per-step failures into opposite-signed policy bias (Section 2).
- **Empirical.** Across two public corpora ($\approx 85k$ trajectories) we quantify how each policy distorts aggregate and model-level reliability over a predeclared hazard grid, including a case where the predicted bias is small (Sections 3–4).
- **Practice.** Censoring-aware estimation under known and misspecified mechanisms, and a minimum validity record for benchmark reports (Section 5).

2 Identification Under Censoring

Let $Y_i \in \{0, 1\}$ be the semantic task outcome, H_i the horizon in agent decision steps, and $C_i \in \{0, 1\}$ the indicator that run i stays valid long enough for Y_i to be observed. The estimand is $p = \mathbb{E}[Y]$.

Proposition 1 (Assumption-free bounds). Let N be initiated runs, M the censored count, and S the observed successes among uncensored runs. With no assumption whatsoever about the outcomes of censored

runs,

$$S/N \leq p \leq (S + M)/N. \quad (1)$$

Proof. Each censored run contributes an unknown $Y \in \{0, 1\}$; setting all to 0 gives the lower endpoint and all to 1 the upper. $\square$

This is the agent-evaluation instance of a classical worst-case bound (Manski 1990), and it reframes one policy immediately: S/N — invalid-as-failure — is not an estimate of reliability but the lower endpoint of the identified interval. Complete-case reporting $S/(N - M)$ estimates $P(Y = 1 \mid C = 1)$, which equals p only under a non-informative observation assumption.

Observation model. We now assume a constant per-step hazard applied independently of agent behaviour, so that

$$\pi_i = \Pr(C_i = 1 \mid H_i) = s^{H_i}, \quad s = 1 - q, \quad (2)$$

together with $C \perp Y \mid H$ and $\mathbb{E}[s^H] > 0$.

Proposition 2 (Horizon-induced complete-case bias). Under (2), the complete-case induced estimand is $p_{cc} = \mathbb{E}[Y s^H] / \mathbb{E}[s^H] = P(Y=1 \mid C=1)$ and

$$p_{cc} - p = \frac{\text{Cov}(Y, s^H)}{\mathbb{E}[s^H]}. \quad (3)$$

Proof. $\mathbb{E}[YC] = \mathbb{E}[Y \mathbb{E}[C \mid Y, H]] = \mathbb{E}[Y s^H]$ and $\mathbb{E}[C] = \mathbb{E}[s^H]$; subtract p and use $\mathbb{E}[Y s^H] - \mathbb{E}[Y] \mathbb{E}[s^H] = \text{Cov}(Y, s^H)$. $\square$

Since s^H decreases in H , the sign of the complete-case bias is the sign of the outcome–horizon association: if successes are shorter, $\text{Cov}(Y, s^H) > 0$ and dropping invalid runs overstates reliability. We stress that p_{cc} is the induced population estimand, i.e. the probability limit of the complete-case ratio estimator, not its exact finite-sample expectation — the expectation of a ratio is not the ratio of expectations. Our Monte Carlo reports the estimator’s realised mean separately.

Proposition 3 (Invalid-as-failure). $\hat{p}_{\text{fail}} = n^{-1} \sum_i C_i Y_i$ is linear in C , so it has exact conditional expectation $n^{-1} \sum_i \pi_i Y_i$, and

$$\mathbb{E}[\hat{p}_{\text{fail}}] - p = \mathbb{E}[Y(s^H - 1)] \leq 0. \quad (4)$$

Equivalently $p - \mathbb{E}[\hat{p}_{\text{fail}}] = \mathbb{E}[Y(1 - s^H)] \geq 0$: scoring invalid runs as failures can never overstate p . Decomposing,

$$\mathbb{E}[\hat{p}_{\text{fail}}] - p = \underbrace{-p(1 - \mathbb{E}[s^H])}_{\text{base rate} \times \text{censored mass}} + \underbrace{\text{Cov}(Y, s^H)}_{\text{association}}, \quad (5)$$

so this bias depends on both the base rate and overall survival and the outcome–horizon association. The first term alone is not a bound on the magnitude: when successes are shorter, $\text{Cov}(Y, s^H) > 0$ partly offsets it, and $|\mathbb{E}[\hat{p}_{\text{fail}}] - p|$ is then smaller than $p(1 - \mathbb{E}[s^H])$.

Proposition 4 (Weighted identification). Known π . If the true $\pi_i > 0$ are known and censoring is generated by the declared mechanism, the Horvitz–Thompson estimator $\hat{p}_{\text{HT}} = n^{-1} \sum_i C_i Y_i / \pi_i$ is exactly unbiased conditional on the finite population, since $\mathbb{E}[C_i / \pi_i] = 1$ (Horvitz and Thompson 1952; Robins, Rotnitzky, and Zhao 1994). This is the regime of our Monte Carlo, where π_i holds by construction. Estimated π . With $\hat{\pi}_i$ fitted from data, exact unbiasedness no longer holds: plugging in an estimate makes the weights random and correlated with the outcome. Under a correctly specified and consistently estimated censoring model, positivity (π_i bounded away from 0), and standard regularity conditions, the plug-in IPCW estimator is instead consistent and asymptotically valid (Robins, Rotnitzky, and Zhao 1994). The Hájek form $\hat{p}_H = \sum_i C_i Y_i / \pi_i / \sum_i C_i / \pi_i$ is a ratio estimator in either regime: consistent and approximately unbiased, not exactly unbiased. All variants degrade as $\pi_i \rightarrow 0$, a positivity problem weighting cannot fix.

What a real deployment would have to estimate. Our $\pi_i = s^{H_i}$ uses each run’s completed horizon, available only because censoring is injected into finished trajectories; a genuinely censored run never reveals its counterfactual completed H . In practice the hazard must be modelled sequentially from information available before censoring (elapsed steps, tokens or wall-clock consumed, quota state). This is the main gap between our sensitivity analysis and a deployed correction.

Three levels of reporting. Together these give a ladder. (L1) No assumptions: report (1). (L2) Non-informative observation: the complete-case reading may be justified. (L3) Modelled horizon-dependent observation: weight or run a sensitivity analysis. IPCW does not solve arbitrary unknown or outcome-dependent censoring, and we do not claim it does.

3 Setup

Corpora. Primary: 80,036 SWE-agent (Yang et al. 2024) trajectories (Nebius 2024) (CC-BY-4.0) on SWE-bench-style tasks (Jimenez et al. 2024; OpenAI 2024), each with a ground-truth resolution label from test execution; we read five pruned columns (≈ 0.6 MB of a 1.11 GB corpus). Secondary: 4,950 τ -bench trajectories (AgentSuite 2026; Yao et al. 2025) from 30 per-model files (29 distinct model identifiers, since one identifier appears in two files), with 165 tasks per file and strictly binary score. We retain every labelled row. In the SWE corpus one row has $H = 0$ (context exhausted before the agent acted); it is kept, since $\pi = s^0 = 1$ makes it never censored. We do not filter on the datasets’ own exit statuses, which are associated with success (resolve rate 0.248 for submitted vs. 0.000 for exit_context) and are part of the historical generating process, not our censoring.

Horizon. H counts agent decision turns only — role=“ai” in the SWE corpus, role=“assistant” in τ -

corpus	N	true p	$\bar{H}$ succ	$\bar{H}$ fail	drop @1%	fail @1%
SWE-agent	80,036	16.73	15.2	28.7	+1.54	-2.27
τ -bench	4,950	55.03	13.1	14.2	+0.23	-6.75

Table 1: Cross-domain summary. Success rates and biases in percentage points at $q = 1\%$. Complete-case bias is governed by the horizon gap alone (13.5 steps vs. 1.1); invalid-as-failure bias combines a base-rate term with the same association term, per (5).

bench — excluding system, simulated-user, and tool turns. Both definitions were fixed before any censoring result. We re-fetched 20 full trajectories per corpus (seed 20260905) and recounted independently: 20/20 exact agreement in both. Our SWE counts are deliberately about half the dataset card’s, which counts environment turns too; $2H+1$ reproduces the card.

Censoring is semi-synthetic; results are a finite-corpus sensitivity analysis. We inject censoring onto completed, labelled corpora. The grid $q \in \{0, 0.10, 0.25, 0.50, 1.00, 2.00\}\%$ per agent step was pre-declared and is a sensitivity analysis, not a measured outage rate. Because these corpora contain repeated tasks and unequal model coverage, we do not treat rows as i.i.d. draws from a universal agent population: our quantities describe what would happen to these corpora under declared mechanisms, and Monte Carlo intervals (200 replicates, seeds from 20260905) reflect injected censoring randomness only, not sampling uncertainty about a wider population.

4 Results

The two corpora dissociate the two biases. Table 1 and Figure 1 are the main result. On SWE-agent, successes are far shorter (15.2 vs. 28.7 steps), so $\text{Cov}(Y, s^H) > 0$ and complete-case reporting inflates reliability: +0.84 points at $q = 0.5\%$ and +2.69 at $q = 2\%$, against -1.20 and -4.12 for invalid-as-failure — a 6.8-point spread between two defensible policies on identical data. On τ -bench the horizon gap is only 1.1 steps, and exactly as (3) predicts the complete-case bias nearly vanishes (+0.40 points at $q = 2\%$). Yet invalid-as-failure is worse there (-12.64 points). By (5) that bias adds a base-rate term $-p(1 - \mathbb{E}[s^H])$ to the same covariance term: on τ -bench the base rate is 55.0% and the association weak, so the first term dominates (-12.94, offset only +0.31), whereas on SWE-agent the base rate is 16.7% and the larger covariance offsets more (-5.87 offset +1.75). Both biases thus involve the association, but only the complete-case bias is governed by it alone — which is why the two corpora dissociate them rather than merely repeating one another.

Bounds are wide. Evaluating (1) at the expected censored count under the declared mechanism (a realised run would give a nearby interval) gives, at $q = 1\%$,

[14.5, 35.3] for SWE-agent and [48.3, 60.9] for τ -bench; at $q = 2\%$, [12.6, 47.7] and [42.4, 65.9]. Reporting a single number without stating the mechanism is a strong assumption, not a neutral default.

The mechanism is visible in horizon strata. On SWE-agent, runs of 1–25 steps are 65.3% of the corpus and succeed at 23.0%, while runs beyond 75 steps are 4.4%, succeed at 2.6%, and retain only 10.5% at $q = 2\%$ against the short stratum’s 77.6%: censoring preferentially deletes long, low-success runs.

IPCW recovers the estimand when the mechanism is known. Horvitz–Thompson bias never exceeds 0.005 points (SWE) or 0.017 points (τ -bench) at any hazard. The cost is variance: HT standard deviation grows from 0.016 to 0.096 points (SWE) as q goes $0.1\% \rightarrow 2\%$. Under a misspecified mechanism (hazard doubling beyond step 25, weighted as if constant) HT bias moves to -0.14 and -0.26 points and Hájek to +0.55 and +0.97 (Appendix A.5): correction degrades gracefully here, but it does degrade.

Ranking distortion is real but rare. With 29 eligible τ -bench models (≥ 100 trajectories; 406 pairs, average ranks for ties) the first pairwise reversal appears at $q = 0.5\%$ under both policies. At $q = 2\%$ complete-case produces 2 reversals ($\rho = 0.998$, $\tau_b = 0.984$) and invalid-as-failure 11 ($\rho = 0.990$, $\tau_b = 0.939$). On SWE-agent, with only 3 eligible models, we observed zero reversals at every hazard — a weak test we report as such. Ordering is fairly robust; margins are not, and comparative claims about gaps are the vulnerable ones.

Absolute reliability is weighting-dependent; the distortion is not. Weighting each task equally rather than each trajectory, the SWE base rate falls materially (16.73% $\rightarrow$ 9.80%) while the $q = 1\%$ biases keep their sign and order of magnitude (+1.14/-1.30); τ -bench moves 55.0% $\rightarrow$ 56.9% with +0.22/-7.29 (Appendix A.4). Absolute reliability is therefore weighting-dependent, but task balancing preserves the direction of both distortions.

5 From Estimand to Practice

Infrastructure invalidation is not hypothetical. In our own instrumented runs, two provider-backed executions completed 9 and 4 agent steps before rate-limit invalidation, and six checkpoint-restored continuations were invalidated before executing any step; no valid outcome existed for any of the eight, and our harness assigned none. This is an existence demonstration only: not a provider reliability estimate, and its frequency is not used as q above.

Minimum validity record. We suggest a benchmark report state, at minimum: (1) initiated runs; (2) runs reaching valid semantic evaluation; (3) invalid/censored counts by reason; (4) the horizon or resource-at-risk distribution; (5) the policy applied to censored outcomes;

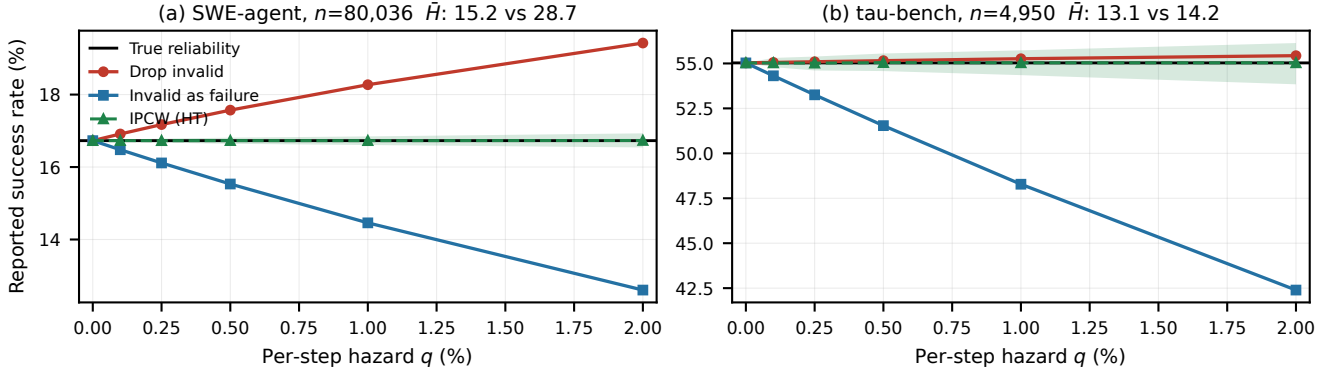


Figure 1: Reported reliability against per-step infrastructure hazard q . This is a synthetic sensitivity analysis: q is a predeclared parameter swept over a grid, not a measured outage rate for any provider. (a) SWE-agent: successes are much shorter, so dropping invalid runs inflates reliability steeply. (b) τ -bench: horizons are nearly outcome-independent, so the complete-case curve is almost flat while invalid-as-failure still falls sharply. IPCW tracks the truth in both (shaded: 95% Monte Carlo interval over 200 replicates, reflecting injected censoring randomness). Note the different y ranges: base rates are 16.7% and 55.0%.

and (6) sensitivity of the headline number to plausible censoring mechanisms. When the mechanism is not defensibly known, report bounds or a sensitivity range rather than silently deleting runs or scoring them as failures. Recording this is cheap; our artifact enforces the underlying separations as regression tests.

6 Related Work

Where prior work studies how agents fail, we study which executions may be counted as evidence. Zhao et al. (2026) anatomise coding-agent failure over complete trajectories; Ruan et al. (2026) abort doomed episodes early, itself a censoring mechanism under our framing whose hazard would need reporting; Li et al. (2026) study failure-aware observability, including runs producing no usable answer, but do not formulate infrastructure termination as a missing-outcome identification problem; Gonuguntla (2026) shows static replay under a substituted model scores a counterfactual world — procedure validity rather than missing outcomes. Two papers use adjacent vocabulary: Wang (2026) uses “censoring” for interface-level output suppression, with no missing-data model, and Raghu, Pandey, and Pandey (2026) extend proper scoring rules to administratively censored trajectories — the closest work, though it scores predictions rather than identifying an aggregate estimand, and uses no inverse-probability weighting. On validity and disclosure, Kapoor et al. (2024), Weidinger et al. (2025), Zhu and Chang (2026), Chen et al. (2026) and Naser Moghadas and Ghaderi (2026) are complementary: disclosure reports what happened, whereas we study how missing outcomes change the estimand. Our estimators are classical (Horvitz and Thompson 1952; Robins, Rotnitzky, and Zhao 1994; Manski 1990); the contribution is the mapping onto agent benchmarks, the horizon-covariance result, and

its magnitude on real trajectories. We do not claim to be first.

7 Limitations

(1) The study is semi-synthetic: we inject censoring rather than observe outages, and the q grid is a sensitivity analysis, not a measured rate. (2) The model assumes independent constant per-step censoring with $C \perp Y \mid H$; the stress test probes one departure, and outcome-dependent (MNAR) censoring is out of scope. (3) IPCW needs a correct, estimable mechanism that real harnesses must fit rather than know, and positivity and variance degrade for long horizons and high q . (4) Both corpora contain repeated tasks and unequal model coverage, so results are finite-corpus statements, not population inference; one τ -bench identifier covers 330 rows because two files share a `model_path`. (5) The SWE ranking analysis has only 3 eligible models and is a weak test, and the real provider case shows existence, not prevalence. (6) Replication on other domains, scaffolds, and genuinely observed infrastructure failures remains future work.

8 Conclusion

An invalid execution is not a failed one. Scoring it as failure and deleting it are biased in opposite directions, and the bias survives even when infrastructure failures are independent of the agent, because horizon compounds exposure. Two corpora behave as the identities predict: where successes are much shorter, deletion inflates reliability; where horizons are nearly outcome-independent, deletion is nearly safe while scoring-as-failure is worst. The implication is concrete: record validity per run, report the censoring mechanism, and when it is unknown report bounds rather than a number.

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A Appendix

A.1 Full hazard grid (SWE-agent). Induced estimands (%), true $p = 16.73$: at $q = 0.10/0.25/0.50/1.00/2.00\%$, $p_{cc} = 16.91/17.17/17.57/18.27/19.42$ and $\mathbb{E}[\hat{p}_{fail}] = 16.48/16.11/15.53/14.46/12.61$, with expected retained fractions $0.974/0.938/0.884/0.791/0.649$.

A.2 Full hazard grid (τ -bench). True $p = 55.03$; $p_{cc} = 55.06/55.09/55.15/55.26/55.43$ and $\mathbb{E}[\hat{p}_{fail}] = 54.31/53.26/51.54/48.28/42.39$; retained $0.986/0.967/0.934/0.874/0.765$.

A.3 Monte Carlo. 200 replicates per hazard, $C_i \sim \text{Bern}(\pi_i)$ with $\pi_i = s^{H_i}$; terminal Bernoulli survival has the same distribution as step-by-step censoring, so the latter is unnecessary. Realised complete-case means agree with the induced estimand to within 0.01 points at every hazard on both corpora, so the ratio-versus-limit distinction is numerically immaterial here even though it matters logically.

A.4 Task-balanced results. Weighting each task equally: SWE-agent $p = 9.80\%$ with $q=1\%$ biases $+1.14/-1.30$ and $q=2\%$ biases $+2.04/-2.36$; τ -bench $p = 56.93\%$ with $+0.22/-7.29$ and $+0.37/-13.61$.

A.5 Reproduction. uv run python scripts/censoring/reproduce_all.py regenerates every number and the figure; no paid API is required. Derived summaries, audits, and provenance (dataset revisions, retrieval dates, licenses) are written to results/censoring/. τ -bench declares no license, so raw trajectories are cached outside the artifact and only derived statistics are stored.

A.6 Prior controlled measurement cases. The artifact also contains controlled demonstrations of four non-censoring invalidity sources: omitting the official evaluator test patch (pytest exits 4, “no tests ran”), pipeline exit-code masking without pipefail, container-wide rather than repository-scoped change attribution ($0 \rightarrow 32$ files from environment churn alone), and future/oracle leakage (20 adversarial future steps alter 0 of 24 prefix features).